Decision Memo: Should we move the first gate from level 30 to level 40?

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Recommendation

Keep the first gate at level 30. Do not launch the level 40 gate. Moving it to level 40 lowered 7-day player retention and showed no benefit on any other measure we checked.

Background

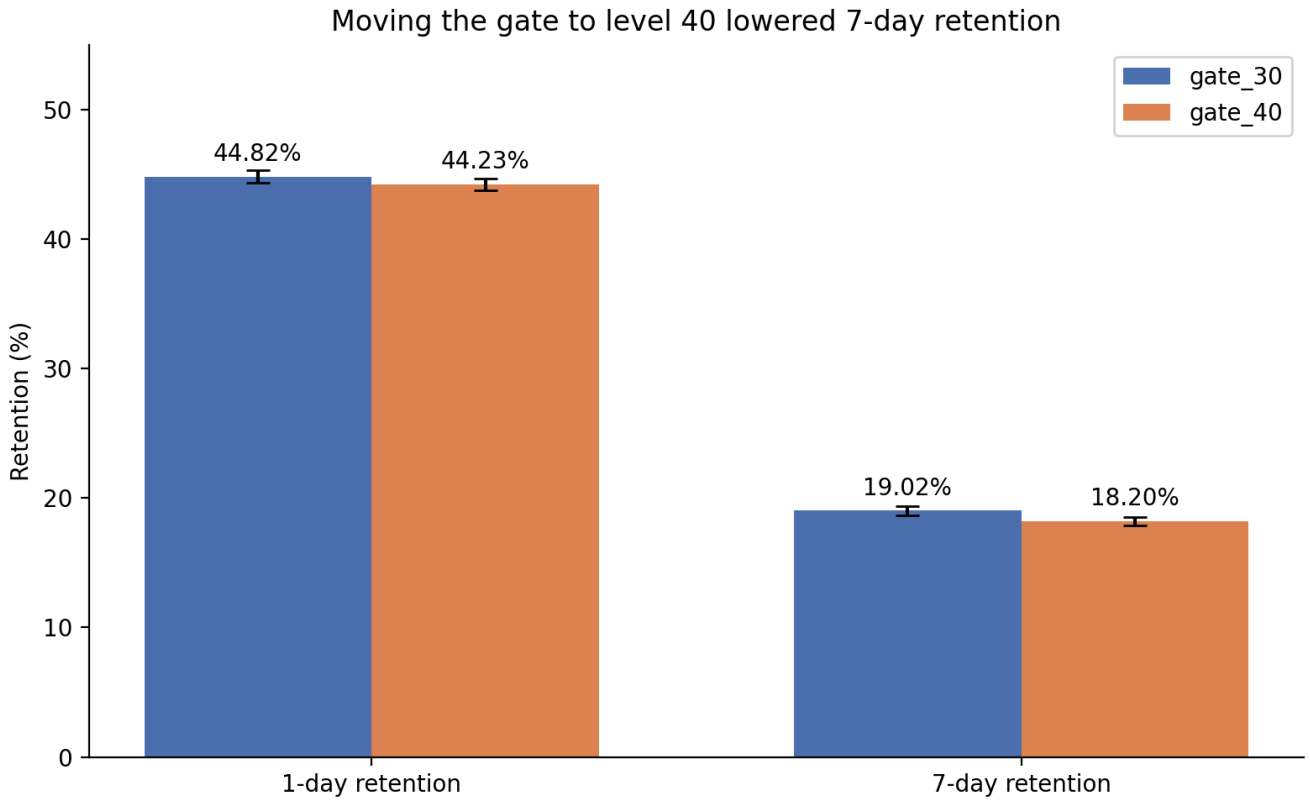
In the mobile game Cookie Cats, players hit a "gate" that forces them to wait or pay before continuing. The team tested moving the first gate from level 30 (control) to level 40 (treatment) to see whether it would keep players engaged for longer.

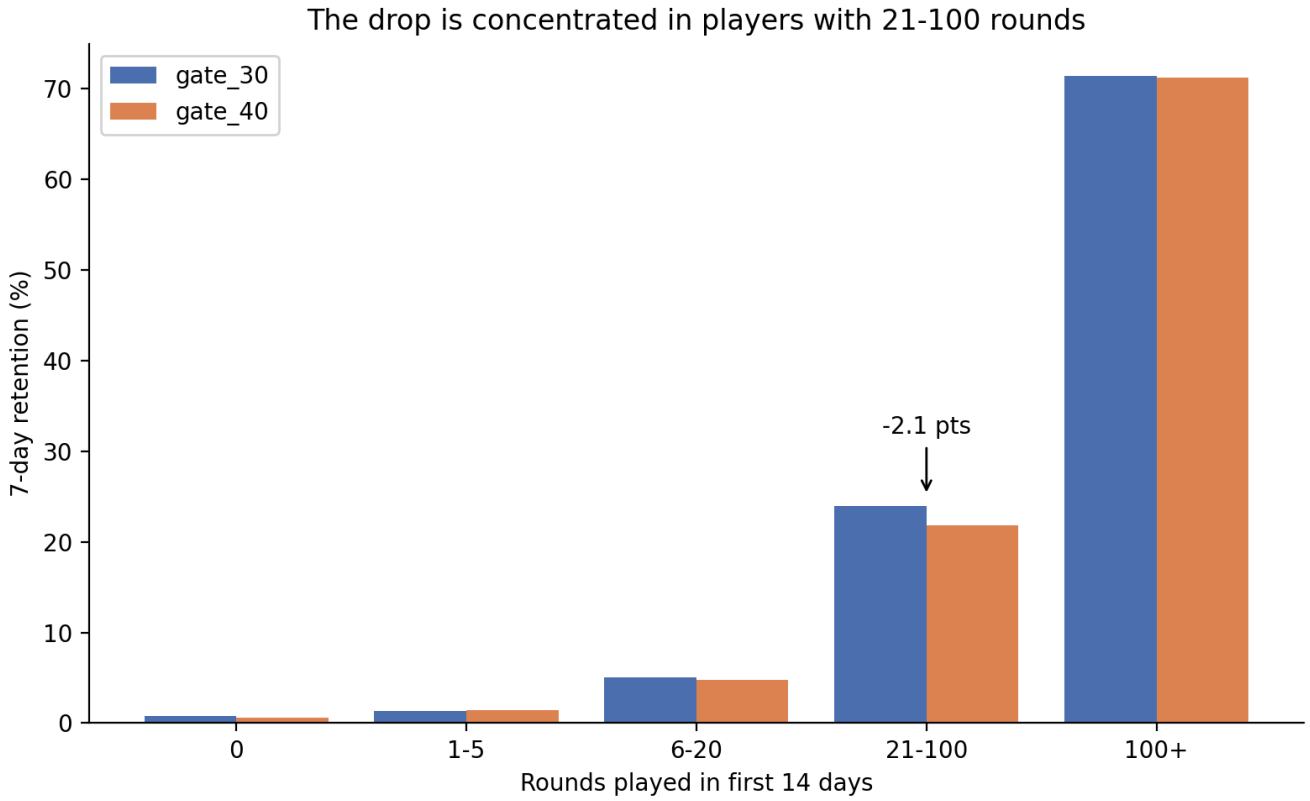
What we tested

- 90,188 players, split between gate_30 (44,699) and gate_40 (45,489). One player with 49,854 rounds was removed as an obvious data outlier.
- Success measures: 1-day retention, 7-day retention (did the player come back), and total game rounds played.

What we found

- **7-day retention fell from 19.02% to 18.20%** (a drop of 0.82 percentage points). The likely range of the true drop is 0.31 to 1.33 points, and the result is statistically significant ($p = 0.0016$).
- **1-day retention** was 44.82% vs 44.23%. This difference is small and not statistically significant ($p = 0.074$), although it leans the same way.
- **Game rounds played** showed no meaningful difference between the two versions (median 17 vs 16).
- The drop is concentrated in **players who played 21 to 100 rounds**, where 7-day retention fell from 23.93% to 21.84% (about 2.1 points). Other player groups showed no clear change.


Moving the gate to level 40 lowered 7-day retention
gate_30
gate_40
Retention (%)
0
10
20
30
40
50
44.82%
44.23%
19.02%
18.20%
1-day retention
7-day retention

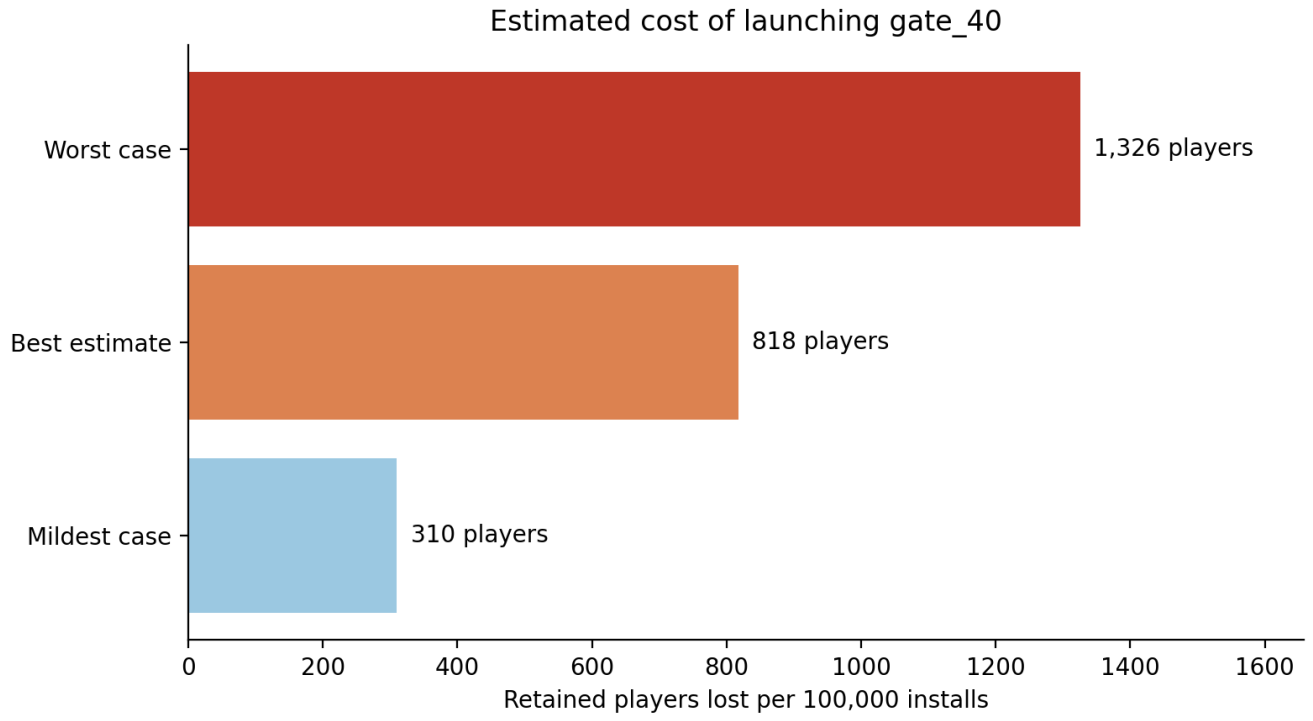

The drop is concentrated in players with 21-100 rounds
gate_30
gate_40
7-day retention (%)
0
10
20
30
40
50
60
70
-2.1 pts
0
1-5
6-20
21-100
100+
Rounds played in first 14 days

Why we trust this result

- Three independent methods point the same way: a standard significance test, a bootstrap resampling check, and a Bayesian analysis (99.9% probability that gate_30 retains better).
- The 7-day result still holds after correcting for running several tests at once (adjusted p = 0.0048).
- The sample is large enough to detect a difference of about 0.74 points, so the test was well powered for an effect of this size.

Business impact (illustrative)

For every 100,000 new installs, launching gate_40 would mean about **818 fewer players still active after 7 days** (range: 310 to 1,326). If each retained player is worth Rs 50, that is roughly **Rs 41,000** in lost player value per 100,000 installs. The Rs 50 figure is an assumption, since no revenue data was available, and should be replaced with the real value per retained player.



Risks and limitations

- The player split between versions was slightly uneven (about 49.6% vs 50.4%) and flagged by a balance check (p = 0.0086). The gap is small, so we do not expect it to change the conclusion.
- Only 7 days of retention were measured, so long-term effects are unknown.
- No revenue or purchase data, so business impact is an estimate built on an assumption.
- The player-group findings use rounds played after assignment, so they show where to look next, not proof of cause.
- The rounds-played result was borderline (p = 0.051), so we can say we found no evidence of a difference, not that none exists.

Decision requested and next steps

1. **Approve keeping the gate at level 30.**
2. If the team still wants to explore later gates, run a follow-up test that measures retention over 14 to 28 days and includes revenue.
3. Consider a targeted test for the 21 to 100 rounds group, since that is where the effect appears.
4. Confirm the real value per retained player so the business impact estimate can be finalised.